# YU LIU

yu130dfb4@gmail.com | China | arXiv:2409.06496

## RESEARCH INTERESTS

Numerical analysis; scientific computing; stochastic differential equations and their numerical methods.

## EDUCATION

**The Chinese University of Hong Kong, Shenzhen** | M.Phil. in Mathematics | 2022–2024

- Cumulative GPA: 3.436/4.000 (official transcript).
- Thesis: Pricing and Trading Strategies of Chinese Convertible Bonds. Advisor: Prof. Gongqiu Zhang.

**Tongji University** | B.S. in Mathematics and Applied Mathematics | 2016–2020

- Major-course average: 85.09/100; overall equivalent average: 82.11/100 (official transcript).
- Thesis: Image Restoration, a Wavelet Frame Based Approach and Beyond. Advisor: Prof. Jae Kyu Choi.
- Selected core coursework (all Grade A): Mathematical Analysis I–III, Real Analysis, Theory of Probability, Numerical Analysis, Computational Method.

## RESEARCH EXPERIENCE

**Monte Carlo Valuation of Chinese Convertible Bonds** | CUHK-Shenzhen | 2023–2024

- Developed and investigated a Monte Carlo valuation framework incorporating path-dependent convertible-bond provisions and backward dynamic programming.
- Used least-squares regression to estimate continuation values; examined a stock-price-regime-dependent regression approach.
- Conducted simulation-based pricing and empirical strategy evaluation, motivating further interest in stochastic modeling and numerical approximation.

**Wavelet-Based Image Restoration** | Tongji University | Undergraduate thesis

- Studied analysis-, synthesis-, and balanced wavelet-frame formulations for image reconstruction.
- Investigated an alternating direction method of multipliers (ADMM) approach for an analysis-based formulation.

## PREPRINT

Liu, Y. and Zhang, G. “Valuation Model of Chinese Convertible Bonds Based on Monte Carlo Simulation.” arXiv:2409.06496 (2024; revised 2026). Preprint.

## TEACHING EXPERIENCE

**Teaching Assistant, MAT1012 Honours Calculus II** | CUHK-Shenzhen | 2024

- Prepared and taught calculus tutorial and exam-review sessions, including proof-based and computational problems for first-year students.
- Graded homework, quizzes, and exams; supported individual problem-solving discussions and received positive feedback on clarity of explanations and student engagement.

## PROFESSIONAL EXPERIENCE

**Backend Software Developer** | BYD, Shenzhen | Sep 2024–Jul 2025

- Developed backend services and workflow automation using Python, Java, Redis, and Go.

**Backend Developer** | Xiaoniu Hulian, Chengdu | Jan–Apr 2026

- Worked on backend development in an enterprise software environment.

## COMPUTATIONAL SKILLS

Python, MATLAB; basic C++ and Java; numerical simulation and computational modeling.